Residence Time Distribution Analysis of CSTR, PFR, and Packed Bed Reactors Using Conductivity Tracer Analysis

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Reactor Selection for Saponification

Chemical reactors are fundamental to industrial chemical processes and manufacturing, where reactor design affects product yield, efficiency, and process cost. In large-scale processes, understanding how fluids move through a reactor is important for maintaining consistent product quality.

The time fluid elements spend in a reactor is called **residence time**, which varies in real systems and is described by **Residence Time Distribution (DTS)**. Deviations from ideal flow influence reaction conversion, mixing, and overall reactor performance.

In this study, DTS was measured for several reactor configurations using pulse injection of a NaCl tracer and conductivity detection at the outlet. Comparing DTS curves for a **CSTR, PFR, and packed bed reactor (PBR)** allows evaluation of how reactor design and operating conditions influence mixing and reactor performance for potential scale-up.

Pilot Reactor System & Experimental Design

The experimental unit consisted of four 0.8 L reactors used to study how reactor design influences residence time distribution (DTS): **Single CSTR, Two CSTRs in series, Tubular Plug Flow Reactor (PFR), and Packed Bed Reactor (PBR).**

For each experiment, deionized water was circulated through the selected reactor at controlled flow rates. A 5 mL of **saturated NaCl solution** was injected as a tracer at the reactor inlet. The tracer concentration at the outlet was monitored using conductivity probe, and measurements were recorded at 2-second intervals until the tracer signal returned to baseline.

The collected conductivity data were later processed to obtain the **residence time distribution function E(t)**, the mean residence time, and the variance, allowing comparison between reactor types and operating conditions.

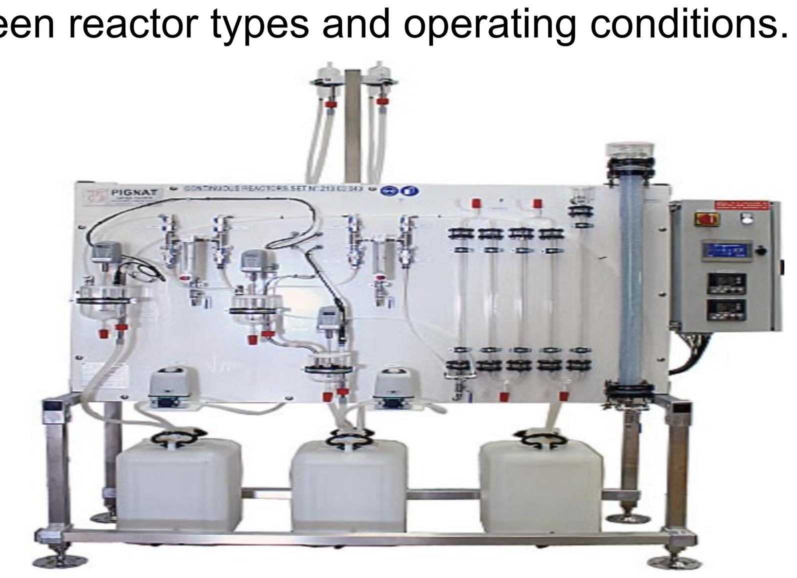

Figure 1: RAP4000 Reactor Unit used for residence time distribution (DTS) experiments.

Reaction Kinetics & RTD Theory

The DTS describes how long fluid elements remain inside a reactor. In real reactors, fluid does not move ideally, so some elements exit earlier or later than the average residence time, affecting mixing, conversion, and reactor efficiency.

The theoretical mean residence time is $\tau = V / Q$ where V is reactor volume and Q is volumetric flow rate.

DTS is measured by pulse injection of a tracer and monitoring outlet concentration over time. The normalized distribution is $E(t) = C(t) / \int C(t)dt$ where C(t) is tracer concentration at time t. The mean residence time and variance quantify the spread of residence times.

Characteristic DTS shapes for each reactor:

- **CSTR**: Broad distribution due to complete mixing
- **Two CSTRs in series**: Narrower distribution approaching plug flow
- **PFR**: Narrow peak near τ
- **PBR**: Near plug flow with slight dispersion

Reactors with **greater axial mixing** will produce broader DTS curves, while **plug-flow systems** will show narrow distributions near τ τ. Increasing flow rate decreases residence time, while higher CSTR impeller speeds increase mixing and broaden the distribution.

Residence Time Distribution Measurements

Table 1. Experimentally determined residence times and modeled conversion values for each reactor under varying operating conditions. Conductivity tracer data were processed to calculate DTS functions, and average residence times were used in reactor performance analysis.

Reactor Type	Condition	Theoretical τ (s)	Experimental τ (s)	Conversion
Single CSTR	200 RPM	288.0	173.8	0.60
Single CSTR	300 RPM	288.0	143.7	0.57
Single CSTR	400 RPM	288.0	122.8	0.55
PFR	10 L/hr	288.0	192.0	0.81
PFR	15 L/hr	192.0	127.0	0.74
PFR	20 L/hr	144.0	95.7	0.68
PBR	10 L/hr	288.0	369.1	0.89
PBR	17 L/hr	169.4	189.3	0.81

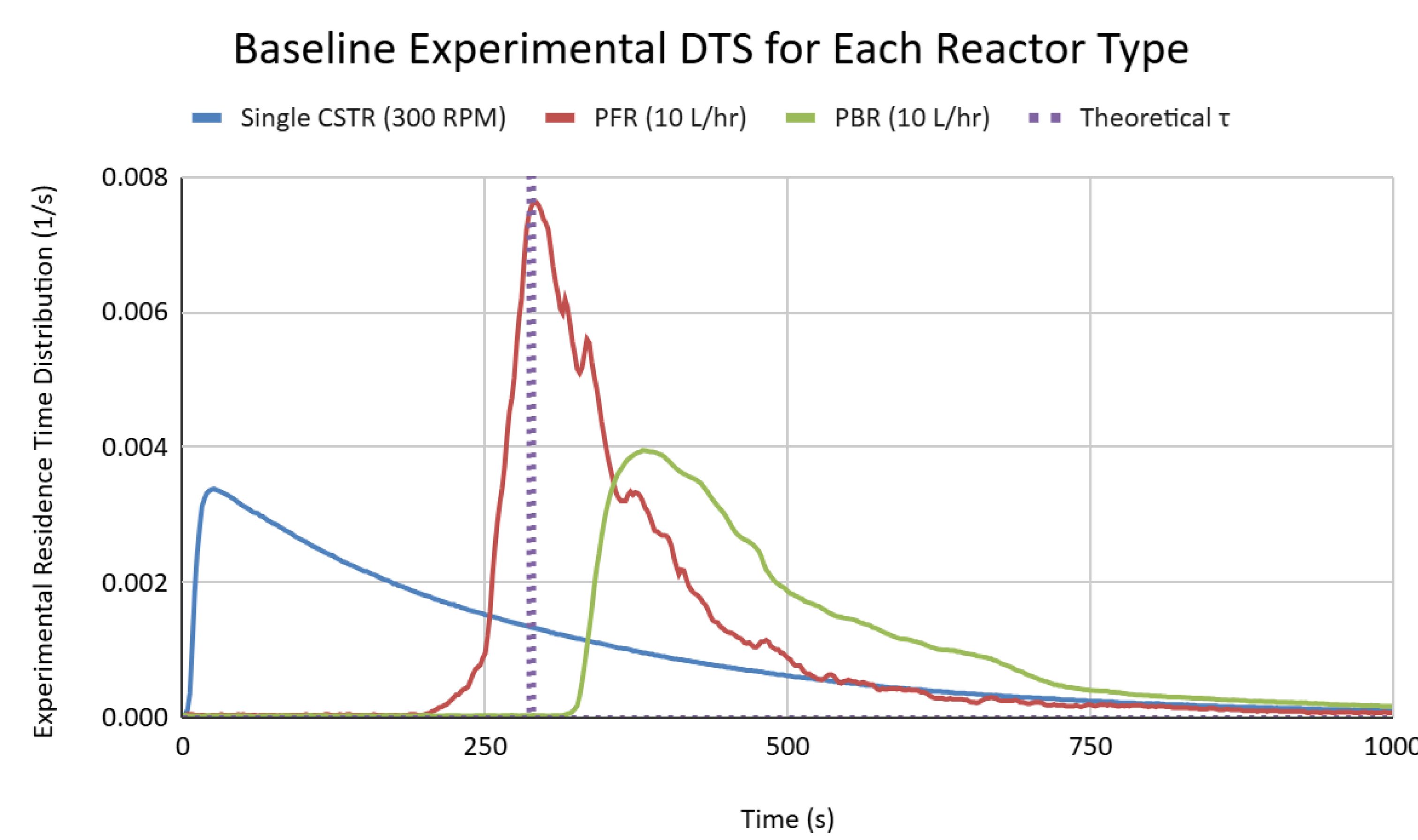

Figure 2. Residence time distribution curves E(t) for the single CSTR, PFR, and PBR at baseline conditions (10 L/hr, 300 RPM for the CSTR). Differences in curve shape illustrate varying degrees of mixing and axial dispersion across reactor types.

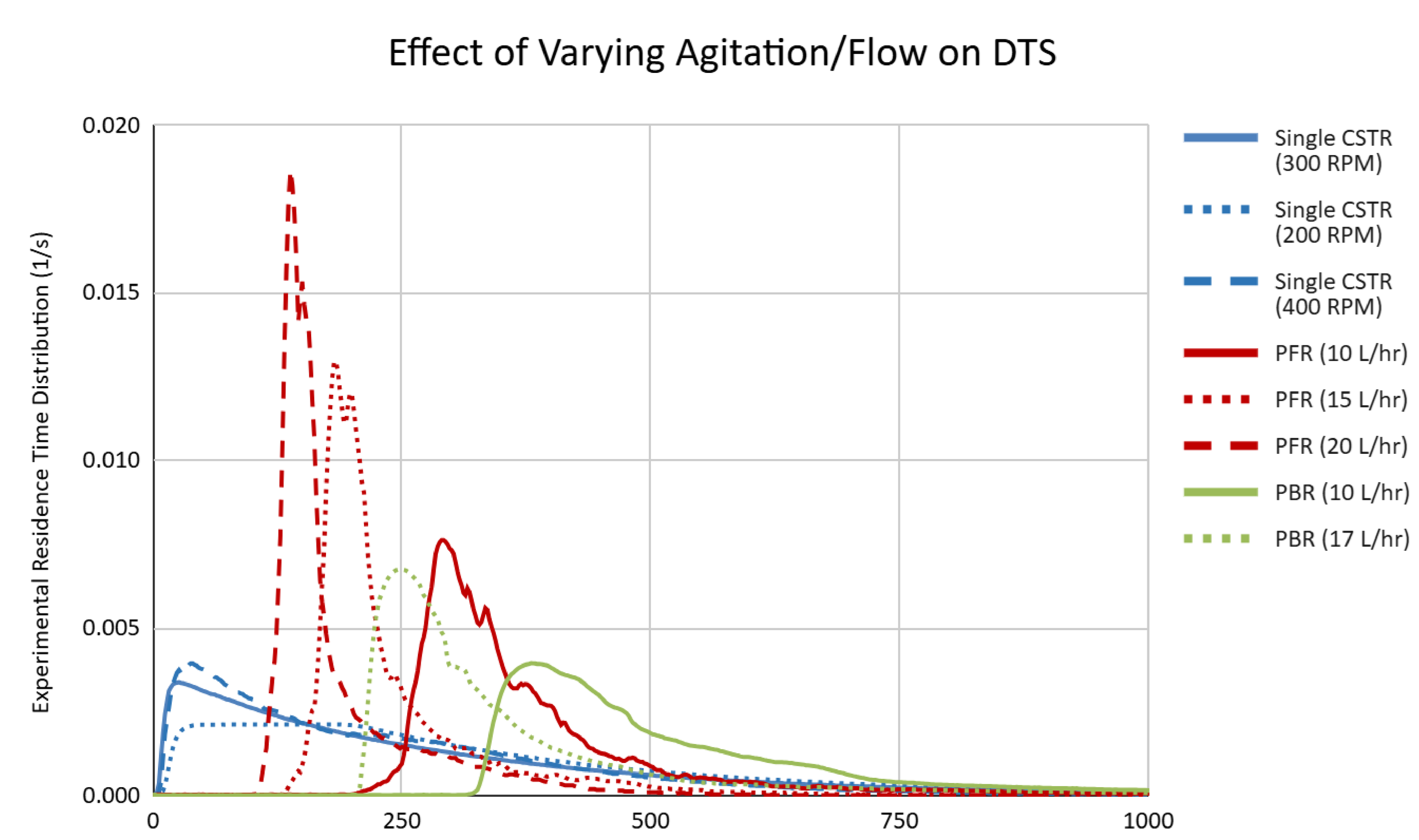

Figure 3. Effect of flow rate and agitation speed on the residence time distribution E(t). Increasing flow rate decreases residence time, while higher CSTR agitation increases mixing and broadens the distribution.

Discussion & Analysis of Findings

The DTS curves showed clear differences in flow behavior across reactor types:

- **CSTR**: The broadest distribution suggests significant **backmixing**. Increasing the agitation speed slightly narrows the DTS suggesting improved mixing, but it **did not** significantly increase **conversion**.
- **PFR**: The narrow peak in DTS curve is consistent with **plug flow behavior.** Increasing the flow rates shift the DTS peak earlier, indicating reduced residence time and **lower conversion.**
- **PBR**: A slightly broader peak than the PFR, with the peak occurring slightly after the expected theoretical DTS, suggests **minor axial dispersion**. Despite this, the PBR still achieved **high conversion** values due to minimal overall backmixing and near plug-flow behavior.

Overall, the results align with theoretical expectations for each reactor type and demonstrate how hydrodynamics influence reactor performance.

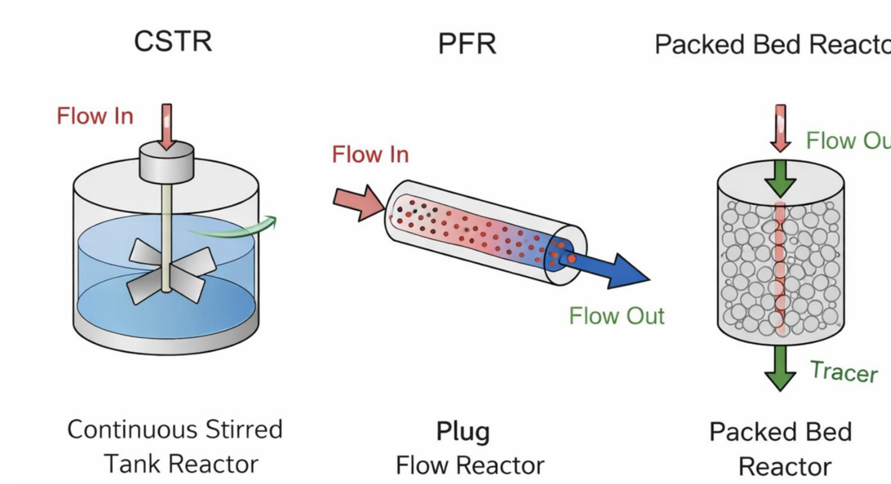

Figure 4. Schematic representation CSTR, PBR, and packed bed reactor configurations.

Operational Recommendations & Scale-up

Based on the DTS analysis, operating conditions should provide sufficient residence time for reaction while maintaining uniform product composition:

- **CSTR**: Agitation speed of **300 RPM** and flow rate of **10 L/hr** is recommended. This condition produced stable mixing and a relatively consistent DTS while avoiding vortex formation.
- **PFR**: A flow rate of **10 L/hr** is recommended to maintain plug-flow behavior and provide sufficient residence time for the saponification reaction. Higher flow rates reduced residence time and lowered conversion.
- **PBR**: A flow rate of **17 L/hr** is recommended. The reactor maintained near plug-flow behavior with high conversion while minimizing axial dispersion at this condition.

For **scale-up of the saponification process**, the **PBR** is recommended due to its high conversion and relatively narrow DTS compared to other reactors. Near plug-flow behavior provides a more uniform residence time and efficient reactant conversion.

Future work could investigate the effect of reactor temperature and catalyst packing characteristics to further optimize efficiency and throughput during scale-up

References

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